Automated Fabric Defect Classification

Roboflow-hosted ResNet18 vs locally fine-tuned PyTorch ResNet18

COMMON 50-IMAGE TEST SET

Roboflow accuracy: 90.00% | PyTorch accuracy: 90.00%

1. Objective and methodology

The objective was to build and compare a hosted image-classification model in Roboflow with a local PyTorch baseline for fabric categories. The Roboflow model used a ResNet18 classification architecture. The custom baseline fine-tuned a pretrained ResNet18 in PyTorch. Both models were evaluated on the same labelled subset of 50 held-out test images.

Item	Recorded configuration
Classes	Defect free, Defected, Waste (Waste retained after instructor clarification)
Initial dataset split	3,499 train / 999 validation / 497 test = 4,995 counted images; 5 fewer than the stated target of 5,000.
Augmentations	Horizontal flip and brightness. Exported training folder contained 10,445 images after augmented training variants were included; validation 999 and test 497.
Preprocessing	Roboflow Version 1, 224 x 224 resize shown in the training settings; classification workflow.
Common comparison subset	50 images: 5 Defect free, 44 Defected, 1 Waste.

Evaluation approach

Roboflow predictions were obtained through its hosted Workflow API and saved to CSV. The local model's saved best checkpoint was evaluated without retraining. The comparison script matched rows by image filename, then computed accuracy and class-averaged precision, recall and F1-score.

Dataset caveat: the 50-image subset is highly imbalanced. In particular, Waste has only one test example, so its class-specific metrics cannot support a reliable general conclusion.

2. Benchmark results

Metric on same 50 images	Roboflow	PyTorch
Correct predictions	45/50	45/50
Accuracy	90.00%	90.00%
Macro precision	81.78%	81.78%
Macro recall	84.39%	84.39%
Macro F1-score	82.93%	82.93%
Mean inference latency	2.45 s/image (client/API round trip)	50.73 ms/image (batched local forward pass)

Interpretation

On the same subset, both systems achieved 90.00% accuracy and made the same predicted class decision on 50/50 images. Each was correct on 45 images and incorrect on 5. The result does not show an accuracy winner on this small sample. Identical class predictions should be treated as an observed result for this subset, not proof that the models are equivalent on other data.

Additional PyTorch full-test result

The saved PyTorch checkpoint was also evaluated on the full exported test folder (497 images): accuracy N/A, macro precision N/A, macro recall N/A, and macro F1 N/A. This result uses a different evaluation size than the 50-image paired comparison and should not be compared directly with Roboflow's 50-image score.

Training time and latency limitations

Roboflow	Roboflow UI estimated 1 h 48 min before training; actual wall-clock duration was not recorded.
PyTorch	Four epochs completed in 101.8 minutes; training was interrupted during Epoch 5. The planned 10 epochs were not completed.
Latency scope	Roboflow's 2.45 seconds includes client-side upload/network/API round trip and remote processing. PyTorch's 50.73 ms is the batched local forward-pass measurement. These are different scopes; they should not be presented as a like-for-like speed test.

Reproducibility note: the PyTorch run completed four epochs and was interrupted during epoch 5 because CPU training was slow. The saved best checkpoint was evaluated, but the planned 10-epoch run was not completed. Roboflow's exact wall-clock training time was not captured; its UI showed an estimate of 1 hour 48 minutes before training.

3. Failure analysis and conclusion

The paired comparison identified 5 misclassified images. Both models made the same five errors: three images labelled Defected were predicted as Defect free, and two images labelled Defect free were predicted as Defected. No Waste error occurred in this subset, but that class was represented by only one image.

Misclassified examples

Image (shortened)	True	Predicted	Roboflow confidence
Defected-1176-_jpg.rf.3c...	Defected	Defect free	91.94%
Defected-1232-_jpg.rf.7d...	Defected	Defect free	89.76%
Defected-1263-_jpg.rf.84...	Defected	Defect free	94.02%
defect-free-15-_jpg.rf.e...	Defect free	Defected	99.86%
defect-free-229-_jpg.rf....	Defect free	Defected	98.50%

Visual review of the contact sheet shows relatively uniform fabric textures at the displayed scale; the defects are not clearly visible in the small previews. Possible explanations include subtle texture differences, ambiguous labels, lighting/shading variation, or limited representation of some visual patterns. These are hypotheses, not confirmed causes: full-resolution image inspection and label verification would be needed to determine the reason. Both models were highly confident on several wrong predictions, so confidence alone did not guarantee correctness.



Conclusion

On the paired 50-image subset, Roboflow and PyTorch both scored 90.00% and had identical predicted labels. This subset is small and imbalanced, so the result is preliminary. The full PyTorch test score was 96.38% on 497 images, but a comparable Roboflow score on the same 497 images was not generated. Future work should use more labelled Waste examples, verify borderline labels, preserve identical test splits, and record actual end-to-end training and inference times.

Final limitations

The 3-class setup follows the instructor's clarification to retain Waste, although the brief initially described two classes. The PyTorch run was stopped before all 10 planned epochs; full training time is therefore not available. No unmeasured timing or causal explanation has been invented.